Technological Complexity Based on Japanese Patent Data

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ABSTRACT

As international competition intensifies in technologies, nations need to identify key technologies to foster innovation. However, the identification is challenging due to the independent and inherently complex nature of technologies. Traditionally, regional analyses of technological portfolios have been limited to binary evaluations, indicating merely whether a region specializes in a technology, or relying on the average Technological Complexity Index (TCI) of the specialized technologies. This study proposes that evaluating TCI at the corporate level could provide finer granularity and more detailed insights. To address the underutilization of corporate-level TCI assessments in Japan, this study applies the Technological Complexity Index using carefully processed patent data spanning fiscal years 1981 to 2010. Specifically, we analyze a bipartite network composed of 1,938 corporations connected to technological fields categorized into either 35 or 124 classifications. Our findings quantitatively characterize the ubiquity and sophistication of each technological field, reveal detailed technological trends reflecting broader societal contexts, and demonstrate methodological stability even when employing finer technological classifications. Additionally, our corporate-level approach allows consistent comparisons across different regions and technological fields, clarifying regional advantages in specific technologies. This refined analytical framework offers policymakers and researchers robust, targeted insights, thereby significantly contributing to innovation strategy formulation in Japan.

Introduction

The intensification of global competition in the twenty-first century has compelled nations to design industrial policies that foster innovation and sustain economic competitiveness. A persistent challenge within these efforts is the identification of the key technologies and actors that form the bedrock of innovative capacity. This difficulty stems from the nature of innovation itself as an intangible phenomenon emerging from a complex, interconnected ecosystem of countries, cities, corporations, and technologies. Effective policymaking therefore requires methodologies capable of pinpointing the critical elements within this web of intricate interdependencies. Yet, a clear, data-driven understanding of which domestic corporations possess the rare and sophisticated capabilities needed to drive growth remains elusive. This study confronts this issue by quantifying technological complexity at the corporate level, drawing on 2.6 million Japanese patent records to identify these pivotal technologies and firms.

Globally, this challenge has been approached from multiple angles. In the European Union, the concept of Smart Specialisation has guided policy support toward key domains[1,2], while Japan's science and technology strategies continue to evolve[3]. A range of indicators has been developed to assess innovation across diverse sectors, from manufacturing to advanced materials[4,5]. Correspondingly, analyses have been conducted at the national[2,6], regional[7–10], and corporate levels[11–13], each contributing to a broader picture of modern innovation ecosystems.

Among these varied approaches, a notable stream of research has emerged from the application of complexity science, particularly the Hidalgo-Hausmann (HH) algorithm[14]. Originally developed to quantify product sophistication using network analysis of international trade data, the algorithm has proven instrumental in highlighting technologies and capabilities of systemic importance. Its core insight is that the true value of a technology cannot be understood in isolation; rather, importance arises from the interplay between how rarely a capability is mastered and how it interconnects with other capabilities across diverse portfolios. This perspective has led to adaptations of the method for analyzing patents and other indicators of technological advancement[15–19].

Building on this foundation, Balland and Rigby developed the Technological Complexity Index (TCI) to measure the complexity of technological fields using patent data[20]. The TCI captures the nuanced relationship between a technology's ubiquity and the sophistication of the actors who possess it[7]. While various extensions and complementary metrics have been proposed[21–23], a comprehensive application of these methods at the corporate level, particularly within the Japanese context, remains a significant gap in the literature. Although Japan has experienced prolonged economic stagnation, its government still struggles to effectively identify the technologies that will drive renewed innovation and international competitiveness[24]. Existing studies on related topics, such as the analysis of automotive suppliers[13], have not provided the broad, complexity-based analysis needed to illuminate the entire landscape of Japanese technology.

This study addresses the aforementioned gap by applying the HH algorithm to patent data from the Patent Office of Japan. We construct a bipartite network linking domestic corporations to technological domains, which is then projected into a monopartite network to derive the TCI. Represented by the second largest eigenvector of the network adjacency matrix, the TCI enables a simultaneous evaluation of technological spillovers and rarity, offering a more robust depiction of the technological landscape than conventional patent counts. Through an analysis of temporal trends from 1981 to 2010, our work provides a detailed assessment of technological capabilities and their evolution, contributing to a deeper understanding of the innovation ecosystem. The insights obtained herein are intended to inform the development of more effective innovation policies and strategies by identifying key technological fields through the lens of complexity.

The remainder of this paper is organized as follows. The Results section presents our findings on the ubiquity and complexity of technological fields and analyzes their temporal trends. In the Discussion section, we provide a critical analysis of our findings and their broader implications. Finally, the Methods section details the data processing and analytical procedures employed in our analysis.

Results

Characteristics of Technologies

In this study, we use the HH algorithm to quantify the structural characteristics of technologies based on patent publications from 1,938 corporations filed between fiscal years 1981 and 2010 (see Methods section). First, we define ubiquity (or degree centrality) $K_{T,0}$ as the number of corporations connected to a technology which lower value indicates that the technology is more rare. Similarly, the average diversity $K_{T,1}$, which measures the variety of corporations connected to a technology, implies that a higher value reflects diversified publication, whereas a lower value suggests specialization. Generally, the average of each indicator is used as a threshold to distinguish high from low values[14,25].

Another key indicator is TCI, which represents the sophistication of a technology by reflecting how advanced the knowledge required for its production. A threshold of zero divides TCI into high and low categories: a high TCI implies that a technology demands a complex combination of unique inputs and advanced technical expertise which indicates only a few advanced corporations can publish it competitively, whereas a low TCI indicates reliance on widely available capabilities, allowing many corporations to participate.

To systematically evaluate technological characteristics, we map the relationships among ubiquity $K_{T,0}$, average diversity $K_{T,1}$, and TCI across patent technologies. (Fig. 1) Technologies are classified according to the Schmoch system[26], as illustrated in Panel A of Fig. 1. Values for ubiquity and average diversity are normalized by their respective means, while TCI is segmented by the zero threshold.

We find a negligible correlation between ubiquity $K_{T,0}$ and average diversity $K_{T,1}$ (Pearson correlation coefficient $r = 0.064$), indicating that widely disseminated technologies are not necessarily associated with diversified corporate portfolios (Panel B in Fig. 1). Technologies within Electrical Engineering, for example, typically show low ubiquity and low average diversity, indicating concentrated innovation within narrowly defined technological domains.

Conversely, ubiquity $K_{T,0}$ and TCI exhibit a moderate positive correlation (Pearson correlation coefficient $r = 0.594$), suggesting that technologies associated with fewer corporations often require complex capabilities (Panel C in Fig.1). In this context, Electrical Engineering exhibits low ubiquity, low average diversity, and low TCI values, indicating that corporations in this field tend to concentrate their innovation efforts within narrowly defined domains.

Detailed analysis using International Patent Classification (IPC) classes further supports these observations (Fig. A1). Within the five major sectors defined by Schmoch[26], the relationship between ubiquity and average diversity is predominantly weak or moderate. Specifically, Electrical Engineering displays a strong positive correlation between ubiquity and TCI (Pearson correlation coefficient $r = 0.738$), highlighting the coexistence of corporations specializing in broadly applicable technologies and those focusing on niche, specialized innovations.

In contrast to Electrical engineering technologies, Chemistry and Pharmaceuticals technologies mostly exhibit high TCI and the high average diversity (Panel D in Fig. 1, Pearson correlation coeffecience $r = 0.316$). These sectors are characterized by rarity and high sophistication and are typically developed by more diversified corporations. Moreover, corporations in these domains are connected not only to fields with low ubiquity but also to those that are more common (e.g., Biotechnology, Analysis of Biological Materials, and Measurement as shown in the right quadrant of Panel C in Fig. 1). A detailed examination reveals a slight negative correlation between ubiquity and average diversity (Fig. A1), suggesting that in sectors with high technological complexity, corporations with more comprehensive technological portfolios tend to produce rare technologies. Furthermore, the IPC classes corresponding to Biotechnology and Measurement, show high TCI values in fields related to the production of sugar and yeast (C13, A21, C12), drug discovery (A61), and the preservation and management of these technologies (C07, A23). These findings are consistent with the socio-technical context of the period, particularly reflecting the emergence of drug discovery technologies in the 1980s and 1990s[27,28]. Moreover, the significant roles of food chemistry and biotechnology during the 1990s and 2000s underscore their contribution to the development of additives that enabled safer food preservation and processing not only in Japan but globally[29,30].

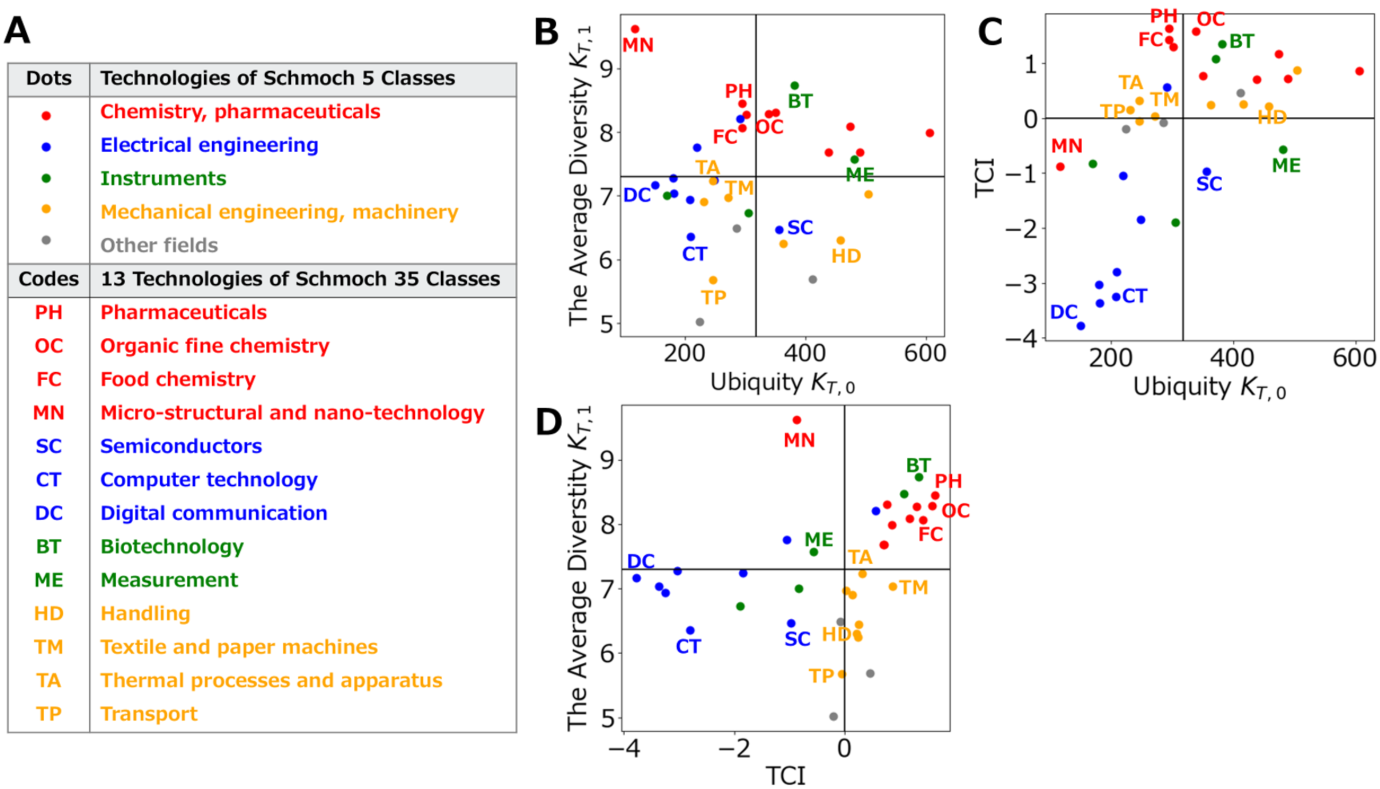

Figure 1. (A) The legend of the five classifications defined by Schmoch[26], along with an explanation of the technology codes used in (B), (C), and (D). (B) Pearson correlation between $K_{T,0}$ and $K_{T,1}$ ($r = 0.064$). (C) Pearson correlation between $K_{T,0}$ and TCI ($r = 0.594$). (D) Pearson correlation between TCI and $K_{T,1}$ ($r = 0.316$). The black lines indicate the mean values of $K_{T,0}$ and $K_{T,1}$, and TCI $= 0$ is also shown with a black line. Each dot is color-coded according to (A).

Finer Technological Trends and Consistency

The traditional regional approach measures technological complexity as a relative index reflecting regional technological capabilities, shaped by the economic advantages arising when corporations cluster within particular countries or cities, known as agglomeration economies[31]. Consequently, this method inherently captures spatial inequalities and is strongly influenced by urbanization levels in Japan. In terms of evaluating a region's technological portfolio, this approach typically provides limited insights, either indicating binary specialization in technologies (0 or 1) or offering only the average TCI of the specialized technologies. Furthermore, finer classifications, such as IPC classes, tend to produce less consistent complexity indices compared to broader classifications like the Schmoch system[32]. Such inconsistencies appear in both regional and technological complexity evaluations due to the symmetric nature of the bipartite network structure underlying these calculations.

In this study, we propose examining technological complexity at the corporate level to overcome these limitations and achieve a more detailed understanding of regional technological portfolios. Specifically, by calculating TCI at the corporate level, we explore whether it is possible to capture more granular insights into technological capabilities within regions. By comparing TCI results derived from regional and corporate approaches, we assess how classification granularity influences the evaluation of inequality and stability of technological complexity across regions and corporations (Panel A in Fig. 2). In this study, we address these inconsistencies by comparing TCI results obtained from regional and corporate approaches, analyzing how classification granularity affects inequality and stability of complexity across regions and corporations (Panel A in Fig. 2).

The scaled TCI of the Schmoch system shows that in the regional approach 26 out of 35 classes fall in the range of 75 to 100. In the corporate approach, only 15 classes lie in that range, whereas the other classes display greater differences among technologies in the regional approach. This difference in stems from the corporate approach being able to characterize each technology in a more detailed manner. The HH algorithm defines ubiquity as the degree of a technology node and this measure depends on the number of paired nodes. The regional approach comprised roughly a few hundred region nodes, such as countries or EU cities[15,33]. The corporate approach has corporation nodes numbering from one thousand to several ten thousand. The iterative calculation of TCI can therefore reflect a wider range of ubiquity.

The difference between the approaches also appears when using a finer classification. When the classification becomes more detailed and the number of technology nodes increases, the regional approach cannot define ubiquity sufficiently. In the case of Civil engineering the variance of TCI values for IPC classes corresponding to the same Schmoch class is larger than that for the corporate approach. The larger difference between the TCI values of the coarse classification and those of the finer classification indicates a greater inconsistency in the evaluation for the regional approach. A Wilcoxon signed rank test shows that the distribution of residuals between the TCI values of the Schmoch and IPC classes is significantly different (p value = 7.23e-14) and that the residuals for the corporate approach are smaller than those for the regional approach (Panel B in Fig. 2).

These consistent and detailed trends are observed not only over a 30 year period but also in evaluations over shorter periods such as 5 years. For example, in the IPC classes corresponding to Food chemistry, the trend of high TCI is consistent in the 2000s. Technologies with high scores as C13 and A21 and those that decline as A23 and A01 are captured in a consistent manner (SI Fig. A2).

Moreover, the corporate approach facilitates relative technological evaluations within specific regions, a capability absent in the traditional regional approach, which relies solely on the national-level presence of technologies. The corporate method identifies strategically important technologies within individual regions. For example, comparative analyses of TCI values for

IPC classes among three representative prefectures, Tokyo, Osaka, and Aichi, highlight distinct local technological strengths. Aichi, notable for its concentration of major manufacturing corporations including Toyota, exhibits a technological profile markedly different from Tokyo and Osaka (Fig. 3). Technologies showing relatively higher local complexity despite lower overall national rankings underline region-specific corporate advantages.

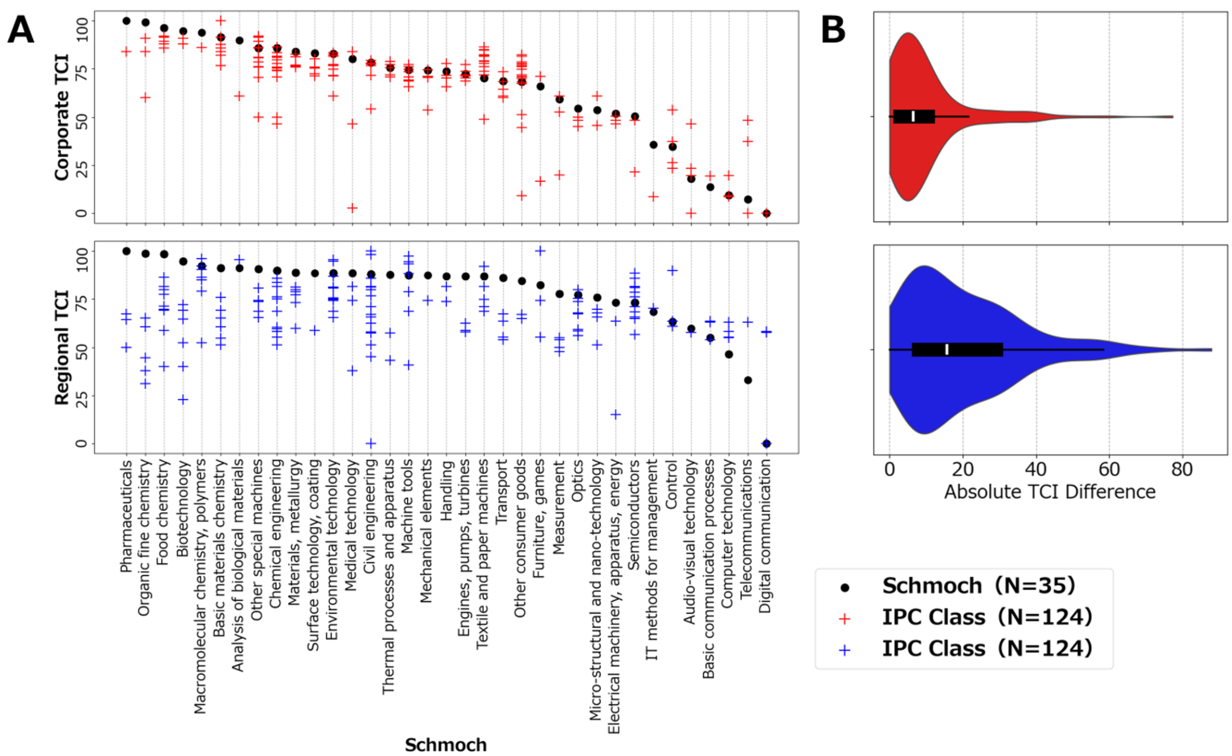

Figure 2. (A) The TCI scores of IPC first Class (N=124) and Schmoch (N=35) for both prefectures and corporations. The scores have been scaled to values ranging from 0 to 100. Each IPC is plotted against the corresponding Schmoch axis, and in cases of multiple correspondences (i.e.,), each pair is plotted individually. (B) The distribution of the absolute differences between TCI of IPC and of Schmoch systems.

Discussion

In this study, we apply the TCI to Japanese corporate patent data, targeting corporations and all patent technological fields, to identify sophisticated technological areas that are key to fostering innovation. Our findings highlight high complexity in the Pharmaceuticals and Chemistry sectors, a result that may reflect the societal conditions of the period, often described as the dawn of the pharmaceutical industry in Japan. Importantly, the corporate level TCI preserves the same relative ranking of technologies even after switching from coarse Schmoch fields to finer IPC classes, demonstrating measurement invariance in contrast to earlier regional studies in which this ordering changed. Although previous works argue that evaluation results of these technological fields may converge because of their classification methods[15], the corporate level approach produced consistent evaluations even with finer classifications. This finding contrasts with the traditional regional approach, which often relies on coarse classifications to keep stability. The ability of the corporate approach to give consistent results at finer

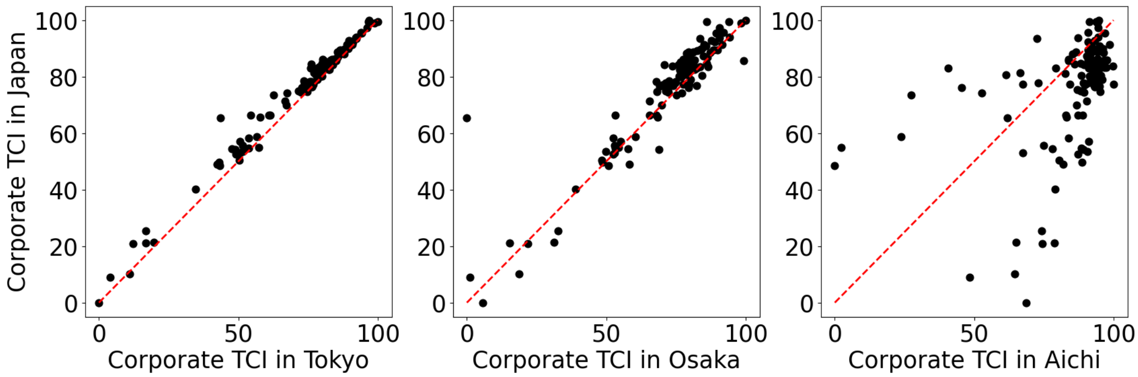

Figure 3. Three scatter plots arranged in a single row compare the Corporate TCI in Japan (vertical axis) to the Corporate TCI in Tokyo (left panel), Osaka (middle panel), and Aichi (right panel) on the horizontal axis. Each axis is scaled from 0 to 100, and the black points represent data points for each classification or entity. The red dashed line indicates the diagonal (y = x).

granularity suggests its potential to provide more accurate and detailed insight into technological trends within specific regions, industries, or other limited scopes.

Limitations specific to the present method remain, particularly because capturing changes in technological trends requires both long term and short term evaluations that detect emerging fields and shifts in innovation patterns, yet such finer grained analyses introduce challenges in data processing. For instance, the calculation of the TCI is sensitive to the number of patents by corporation and technological field in each period. Shortening the aggregation period may introduce corporations with only a few patents, which adds noise to the TCI and reduces the stability of the results[32]. This limitation reflects a shortcoming of the HH algorithm, which evaluates network complexity from node degrees in a bipartite graph and therefore relaxes the conditions for link formation when the data are sparse. Several mathematical and algorithmic improvements can mitigate this issue. The Fitness Complexity algorithm[21] uses nonlinear iterative calculations that better account for network structure and shows strong correlations with indicators based on the HH algorithm while improving robustness in the evaluation of regional and corporate competitiveness[22,23]. In addition, links can be weighted by citation counts instead of raw patent counts to obtain a more systematic and scalable measure of technological effort. Applying these enhancements within our framework would allow deeper insight into Japan's industrial technology landscape. Future research should extend this work in several directions. Robustness checks that compare the corporate level TCI with alternative indicators such as citation counts, methods that predict temporal changes in the bipartite graph using machine learning, and cross national comparisons are promising next steps.

Ultimately, this study represents a cornerstone for understanding the characteristics of domestic industries in Japan through the lens of technological complexity. By identifying key technological areas, the findings offer guidance that can inform industrial policies and strategies, especially during a period marked by rapid progress in artificial intelligence and growing international competition.

Methods

Data

The dataset utilized in this study comprises registered patents submitted to the Japan Patent Office (JPO) covering fiscal years 1981 through 2010, encompassing detailed records of 3,189,536 patents filed by 64,622 corporations. Patent records adhere to the International Patent Classification (IPC) system and are further categorized according to the Schmoch classification[26].

The Schmoch system aggregates IPC codes into 35 distinct technological fields, ensuring balanced class sizes and technological consistency, and has been widely applied in previous research[7,8,10]. Nevertheless, the granularity of classification substantially influences analytical outcomes. Administrative classifications vary markedly in granularity and international comparability, potentially introducing analytical noise or biases if inadequately managed[15]. Employing consistent classification schemes, exemplified by the Schmoch system utilized in this study, mitigates such issues by providing balanced and coherent technological categories for robust analyses.

Traditional methods for measuring patent contributions typically employ raw patent counts[6–10,20,34]. However, such counting methods can lead to overestimation, particularly when patents involve multiple corporations or span multiple technological domains[8]. To address potential overestimations, the primary IPC class of each patent is mapped onto the corresponding Schmoch field, with patent value fractionally allocated by equally dividing it among associated corporations and technological fields. This fractional allocation prevents overestimation arising from multi-corporation ownership or cross-domain classifications.

Additionally, filtering criteria are essential to exclude extreme cases capable of distorting analytical outcomes[8,15]. Accordingly, a filtering criterion based on patent distribution per corporation is applied in this study, retaining only the top 3% of corporations in patent output (Fig. effig patentdistribution). This filtering criterion excludes corporations exhibiting minimal patenting activity, thereby reducing significant analytical noise in network analyses. Following this filtering, the dataset comprises 2,485,027 patents filed by 1,938 corporations, representing approximately 93% of the original patent volume.

Utilizing these filtered data, technological complexity is based on a bipartite network connecting corporations to technological fields in which they are specialized. Although there are some arguments regarding the optimal specialization index for establishing connections within these networks[8], we follows prior research by employing the Revealed Technological Advantage (RTA) index[35]. The RTA index builds upon the revealed comparative advantage framework proposed by Balassa[36]. Specifically,The RTA index builds upon the revealed comparative advantage framework proposed by Balassa[36]. Specifically,the RTA index for a given corporation in a specific technological field measures the ratio between the share of patents held by corporation C in technology T and the share of patents in technology T among all corporations. The numerator of the RTA index captures the fraction of patent activity devoted to technology T for corporation C. Similarly, the denominator reflects the overall share of patent activity in technology T among all corporations, computed as the sum of W_{CT} over all corporations divided by the sum of W_{CT} over all corporations and technologies. The RTA index is thus given by

$$\mathrm{RTA}_{CT} = \frac{W_{CT}}{\sum_T W_{CT}} \bigg/ \frac{\sum_C W_{CT}}{\sum_{CT} W_{CT}} \ . \tag{1}$$

We consider that RTA index values of 1 or greater indicate that corporation C holds a prominent technological advantage in technology T. Based on this threshold, a binary adjacency matrix of the bipartite network M_{CT} is defined by

$$M_{CT} = \begin{cases} 1 & \text{if } \mathrm{RTA}_{CT} \geq 1, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Technological Complexity

The binary matrix M_{CT} forms the two metrics which characterize each node in the bipartite network by the sum of connections, known as degree centrality. The diversity of corporations $K_{C,0}$ quantifies the number of technologies in which a corporation is specialized, and indicated by M_{CT}. Similarly, the ubiquity of technologies $K_{T,0}$ represents the number of corporations that exhibit a technological advantage in a given field. These quantities are computed as

$$K_{C,0} = \sum_T M_{CT}, K_{T,0} = \sum_C M_{CT} \ . \tag{3}$$

Higher diversity values indicate corporations with broad technological capabilities, while lower ubiquity values signify that the technologies are rare. Through calculating these two characteristics of adjacent nodes iteratively, the diversity and ubiquity of each node is updated as

$$K_{C,N} = \frac{1}{K_{C,0}} \sum_T M_{CT} K_{T,N-1}, \tag{4}$$

$$K_{T,N} = \frac{1}{K_{T,0}} \sum_C M_{CT} K_{C,N-1}. \tag{5}$$

Here, the metric $K_{T,1}$ represents the average nearest neighbor degree for technology nodes and it reflects the average diversity of corporations connected to a given technology. Substituting Eq. 4 into Eq. 5 yields

$$K_{T,N} = \sum_{T'} \widetilde{M}_{TT'} K_{T',N-2} \tag{6}$$

where T' denotes a technological field, and $\widetilde{M}_{TT'}$ is defined as

$$\widetilde{M}_{TT'} = \sum_{C} \frac{M_{CT}\, M_{CT'}}{K_{C,0}\, K_{T,0}} \,. \tag{7}$$

Eq. 6 is satisfied when $K_{T,N} = K_{T,N-2} = 1$, which corresponds to the eigenvector associated with the largest eigenvalue of the stochastic matrix $\widetilde{M}_{T,T'}$. Since this eigenvector is a vector of ones and not informative, the eigenvector corresponding to the second largest eigenvalue of $\widetilde{M}_{TT'}$ captures the greatest variance among technology fields[15,37]. Thus, TCI is given by

$$TCI = \frac{\widetilde{T} - \langle \widetilde{T} \rangle}{\mathrm{stdev}(\widetilde{T})}, \tag{8}$$

where $\widetilde{T}$ represents the eigenvector corresponding to the second largest eigenvalue of $\widetilde{M}_{TT'}$, and $\langle \widetilde{T} \rangle$ is their mean and $\mathrm{stdev}(\widetilde{T})$ is their standard deviation.

Data Availability

The patent data used in this study are publicly available from the Japan Patent Office (JPO) database (https://www.j-platpat.inpit.go.jp/). Processed data supporting the findings of this study are available from the corresponding author upon reasonable request.

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Author Contributions Statement

H.I. collected the data and secured the funding. R.K. designed the study, conceived the main research idea, performed the analysis, and drafted the manuscript. Both H.I. and R.K. critically reviewed the results, contributed to the revisions of the manuscript, discussed the findings, and approved the final version of the manuscript.

Additional information

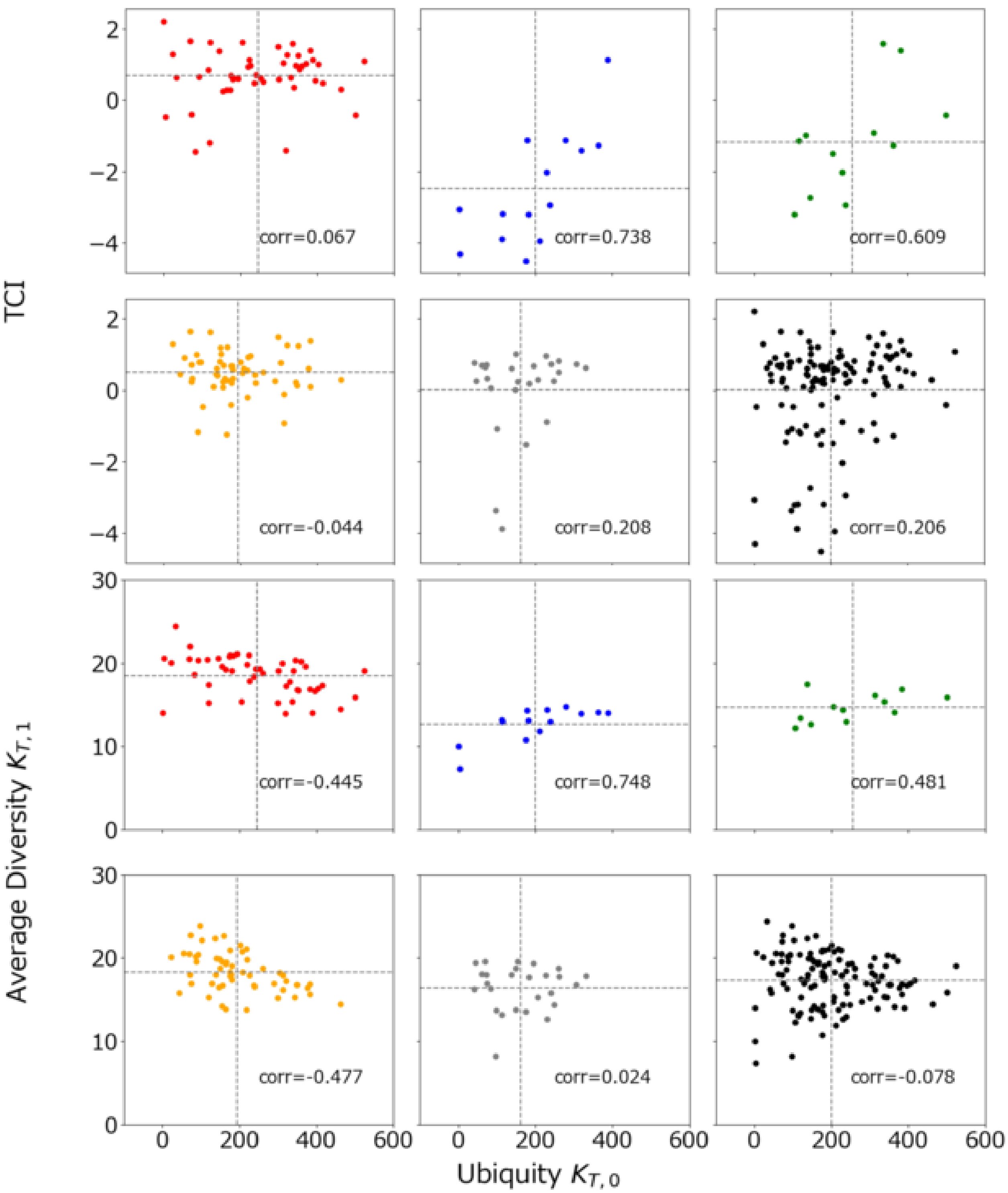

Figure A1. The Pearson correlation between the degree centrality $K_{T,0}$ of technology node T classified by IPC and the average nearest neighbor degree $K_{T,1}$, and the Pearson correlation between the degree centrality $K_{T,0}$ of technology field node T and TCI. In both figures, the black lines indicate the mean values. Each data point is color-coded according to the five classifications defined by Schmoch[26].

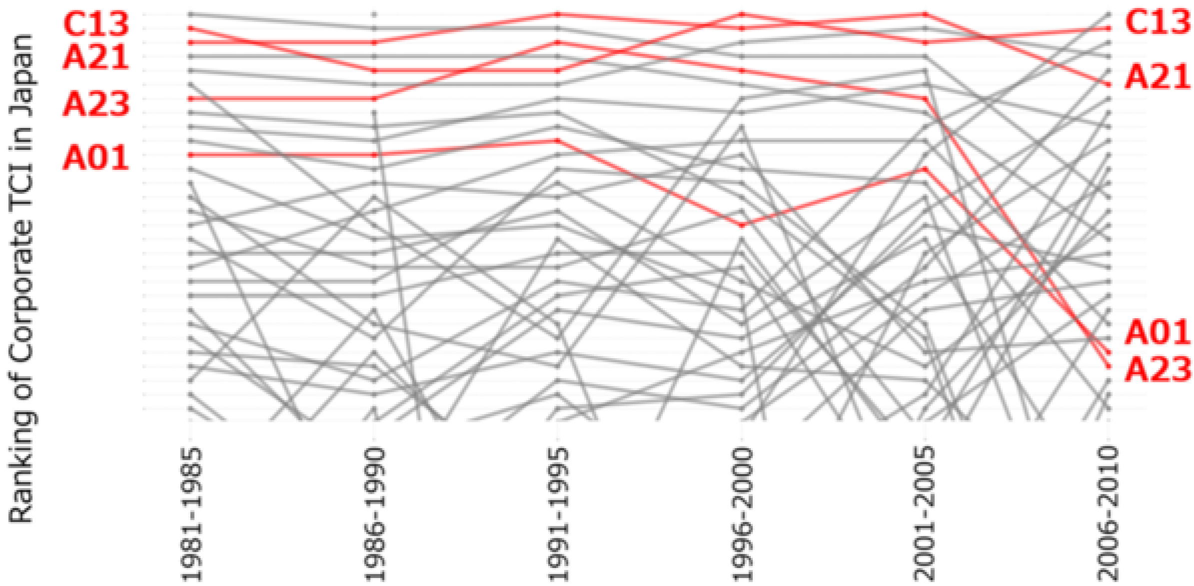

Figure A2. The top 30 TCI rankings in five year increments from 1981 to 2010. IPC classes associated with Food Chemistry (C13, A21, A23, and A01) are highlighted in red.

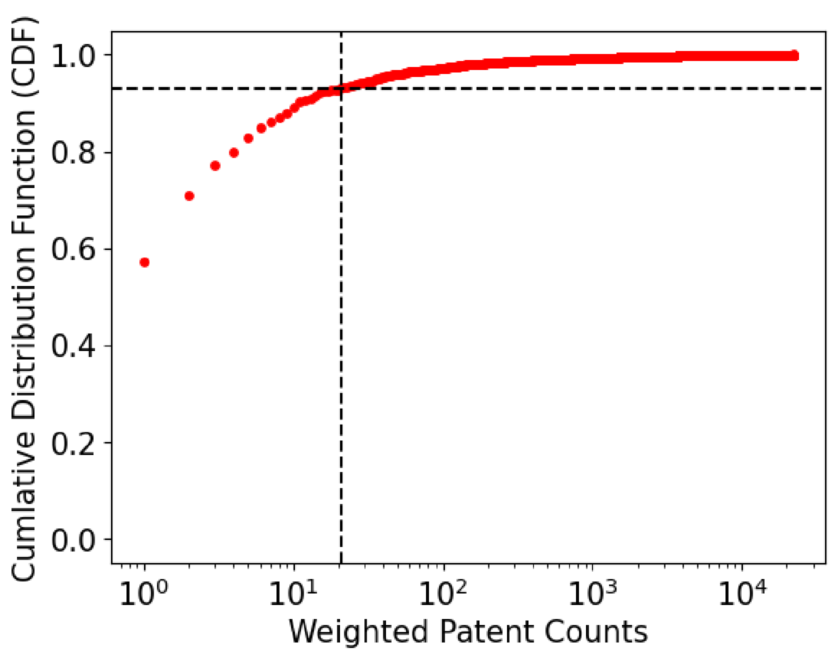

Figure A3. The distribution of weighted patent counts (horizontal axis) for each corporation, with the cumulative distribution function (CDF) of their patents on the vertical axis. The black dashed lines on both axes indicate the excluded region, corresponding to the top 3% of corporations that account for 93% of all patents.